

Session 13 • Thursday 8 October 2026 • The gain derivation is projected; the board carries the log-slope argument, the matching test, and the two bills.

Board notes

Print this and carry it. The deck records what is *projected*; this records what gets *written*.

Three board moments. Why a log slope, at 28. The matching test as four numbers and two inequalities, at 38. And the two bills side by side, at 56, which is the sentence the session exists to produce.

Conventions. **ASK:** put it to the room before writing, and the answer to expect. **POINT:** turn and point at the board rather than the screen. **IF ASKED:** the question that always comes.

Before class — left wing, the gate

Draw it once and leave it up all period. A promoter, a repressor arriving, a protein leaving:

$x \neg [P] \rightarrow y$

and underneath, the curve in one line:

$$y = y_{\min} + \frac{y_{\max} - y_{\min}}{1 + (x/K)^n}$$

Leave the center board empty. It fills at 28.

Right wing, minutes 15–23 — the group answers

Three columns, one per question. Column 1 is the one every group reaches: stage two's steep part sits on stage one's swing. Write their words, then write the chain rule underneath them:

$$\frac{d \ln z}{d \ln x} = \frac{d \ln z}{d \ln y} \cdot \frac{d \ln y}{d \ln x}$$

ASK at 23, before sorting: *what has to be true for that product to be bigger than either factor?* Both factors above 1, at the same input. That is the whole session in one line, and it is better coming from them.

Column 3 is where to spend the time. Hooshangi's own **first** three-stage build failed this way — a mutant λ promoter with an elevated low output, replaced with the original OR1 (p. 3585) — but that was a prototype they discarded, so it shows that leak can ruin a stage rather than explaining the seven percent the shipped circuit 3 delivered. Their candidates for that are on the same page: noise limiting the gain from added elements, and LacI expressed far above what full repression of cl needs.

Center, at 28 — why a log slope

This is the step where the room will nod along without understanding, so make them do it. Write:

$$\text{electronics: } \frac{dV_{\text{out}}}{dV_{\text{in}}} \qquad \text{here: } \frac{d[\text{TetR}]}{d[\text{aTc}]} = ?$$

ASK: *what are the units of the second one?* Molecules of TetR per molecule of aTc — and nobody chose those units for a reason. **Then ASK:** *what happens to that number if I report TetR in nM and you report it in molecules per cell?* It changes by a thousand, and the gate did not.

Write the fix:

$$G = \frac{d \ln y}{d \ln x} = \frac{x}{y} \frac{dy}{dx} \quad (\text{fractional out per fractional in — no units})$$

And the method, because nothing else demonstrates it. $|G| = 1$ does not factor. Evaluate $|G|$ on a grid, find the peak, then bisect on each side. Write those three words on the board — *grid, peak, bisect* — because that is what `handout item 2` asks for and what `posb.digital` does.

IF ASKED whether this is standard: Daniel et al. call the same quantity sensitivity (p. 623), and for a chain of identical gates measured in the same units the two definitions agree about where the restoring region is.

Center, at 32 — the leak, and why it is welcome

At derivation step 5, before the slide shows it:

ASK: *we have been treating leak as a nuisance all term. Here it does something useful — what?* Let them look at the gain curve on the screen. Without y_{min} the gain would stay above 1 forever and there would be no upper threshold, no defined HIGH, and nothing to match against.

Write underneath:

$$|G| = g \cdot \frac{y - y_{\text{min}}}{y} \quad \implies \quad |G| = 1 \text{ has two roots}$$

CHECK: at $n \leq 1$, $g < 1$ everywhere, so there are no roots at all. Point at the left wing: that is the toggle's condition from session 9, and the ring's from session 11, doing a third job.

And the half that is easy to miss: $n > 1$ is necessary, not sufficient. The peak has to clear 1, and a large enough leak stops it. At $n = 2$ a hundredfold swing gives a peak of 1.64, a tenfold swing 1.04, and a fivefold swing 0.76 — no switch at all. Two of Cello's twenty gates are in that class, which is the surface at 44.

Center, at 38 — the matching test

Four numbers and two inequalities. Write them as a box and leave them up for the rest of the session:

$$x_{IL}, x_{IH} \quad (\text{what a gate reads}) \qquad y_{OL}, y_{OH} \quad (\text{what a gate emits})$$

$$\text{NM}_L = \log \frac{x_{IL}(B)}{y_{OL}(A)} > 0, \quad \text{NM}_H = \log \frac{y_{OH}(A)}{x_{IH}(B)} > 0$$

POINT at which letter is A and which is B each time. Getting the sender and the receiver the wrong way round is the commonest error, and it gives a plausible wrong answer rather than an obvious one.

ASK before the RBS slide: *name a change that moves what a gate emits and not what it reads.* Fish for the RBS. Then write the reason, because it is one line and it is the whole argument:

$$y \rightarrow fy \implies \frac{y - y_{\min}}{y} \text{ unchanged} \implies G \text{ unchanged} \implies x_{IL}, x_{IH} \text{ unchanged}$$

Then the caveat, in one line, because the slide carries it: that holds when the output has its own RBS. In a repressor chain the output protein *is* the next gate's repressor, so one RBS sets both, which is why Nielsen's thresholds moved when they changed it.

Center, at 56 — the two bills, and the sentence

Clear a space and write these two lines under each other. They are the point of the session:

$$\text{matching window} \uparrow \text{ with } n \quad \text{log-linear range} = 9^{1/n} \downarrow \text{ with } n$$

Say it, slowly: the same cooperativity that makes a gate easy to wire to other gates is what makes it useless as a measuring instrument. Digital is not free and it is not a property of the biology. It is a regime you pay to operate in.

IF ASKED where $9^{1/n}$ comes from: on a log axis the slope of $u^n/(1+u^n)$ is $ny(1-y)$, largest at $y = 1/2$; staying within 75% of that peak keeps y between $1/4$ and $3/4$, and the input ratio between those two points is $9^{1/n}$. It is on the answer sheet.

What to cut if you are late

In this order, last first:

1. The second Nielsen surface (the numbers). Keep the figure and say which side fails without the arithmetic.
2. The synchrony row on the delay slide.
3. Handout item 4 — but then say the $9^{1/n}$ sentence out loud, because it is the session's thesis and item 4 is where it otherwise lands.

Never cut the matching test (PS6 assesses it), the leak step (it is what makes a threshold exist), or the two bills at 56.